

Environment Variables Reference Guide

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Environment Variables Reference

Overview

This document provides a complete reference for all environment variables used by the ClientIQ application. Variables are organized by functional category.

Configuration File Location

- **Development:** .env in project root
- **Production:** C:\ClientIQ\config\.env

Quick Reference

Category	Required	Variables
Application	Yes	NODE_ENV , PORT
Database	Yes	DATABASE_DIALECT , MSSQL_*
Session	Yes	SESSION_SECRET
SAML	Yes	SAML_*

Application Settings

NODE_ENV

Required: Yes

Default: development

Values: development , production , test

Sets the runtime environment.

```
NODE_ENV=production
```

Effects:

- production : Enables optimizations, disables debug logging
- development : Enables hot reloading, verbose logging
- test : Used for automated testing

Windows Note: The syntax `NODE_ENV=development npm run dev` does not work on Windows. Use one of these alternatives:

```
# PowerShell
$env:NODE_ENV = "development"
npm run dev
```

```
# Command Prompt
set NODE_ENV=development && npm run dev
```

See [Dev/Test Server Setup Guide](#) for complete Windows configuration.

PORT

Required: No

Default: 5000

Type: Integer (1024-65535)

HTTP port for the Node.js application.

```
PORT=5000
```

Note: In production, Nginx proxies to this port. Only bind to localhost.

HOST

Required: No

Default: 0.0.0.0

Network interface to bind to.

```
HOST=0.0.0.0
```

Database Configuration

DATABASE_DIALECT

Required: Yes

Default: postgresql

Values: postgresql, postgres, pg, sqlserver, mssql

Specifies the database backend.

```
DATABASE_DIALECT=sqlserver
```

PostgreSQL Variables

Used when `DATABASE_DIALECT` is `postgresql` :

Variable	Required	Description
<code>DATABASE_URL</code>	Yes	Full connection string

```
DATABASE_URL=postgresql://user:password@host:5432/clientiq
```

SQL Server Variables

Used when `DATABASE_DIALECT` is `sqlserver` :

Variable	Required	Default	Description
<code>MSSQL_SERVER</code>	Yes	-	Server hostname or IP
<code>MSSQL_DATABASE</code>	Yes	-	Database name
<code>MSSQL_USER</code>	Yes	-	SQL Server login
<code>MSSQL_PASSWORD</code>	Yes	-	Login password
<code>MSSQL_PORT</code>	No	1433	TCP port
<code>MSSQL_ENCRYPT</code>	No	true	Enable TLS encryption
<code>MSSQL_TRUST_SERVER_CERTIFICATE</code>	No	false	Trust self-signed certs

```
# SQL Server Configuration
MSSQL_SERVER=sql-server.yourbank.local
MSSQL_DATABASE=ClientIQ
MSSQL_USER=svc_clientiq
MSSQL_PASSWORD=YourSecurePassword123!
MSSQL_PORT=1433
MSSQL_ENCRYPT=true
MSSQL_TRUST_SERVER_CERTIFICATE=false
```

Connection Pool Settings

Variable	Default	Description
DB_POOL_MIN	2	Minimum pool connections
DB_POOL_MAX	10	Maximum pool connections
DB_POOL_IDLE_TIMEOUT	30000	Idle timeout (ms)

```
DB_POOL_MIN=5
DB_POOL_MAX=20
DB_POOL_IDLE_TIMEOUT=60000
```

Session Configuration

SESSION_SECRET

Required: Yes

Type: String (minimum 32 characters recommended)

Secret key for signing session cookies.

```
SESSION_SECRET=your-64-character-random-string-generated-with-crypto
```

Generate a secure secret:

```
# PowerShell
[System.Web.Security.Membership]::GeneratePassword(64,10)

# Or using Node.js
node -e "console.log(require('crypto').randomBytes(32).toString('hex'))"
```

SESSION_TIMEOUT

Required: No

Default: 3600000 (1 hour)

Type: Integer (milliseconds)

Session idle timeout.

```
SESSION_TIMEOUT=1800000 # 30 minutes
```

SESSION_COOKIE_NAME

Required: No

Default: clientiq.sid

Name of the session cookie.

```
SESSION_COOKIE_NAME=clientiq.session
```

SESSION_COOKIE_SECURE

Required: No
Default: true in production

Set to true in production (requires HTTPS).

```
SESSION_COOKIE_SECURE=true
```

SAML Configuration

Core SAML Settings

Variable	Required	Description
SAML_ENTRYPOINT	Yes	IdP SSO URL
SAML_ISSUER	Yes	SP Entity ID
SAML_CALLBACK_URL	Yes	ACS endpoint URL
SAML_CERT	Yes	IdP certificate (base64)

```
# SAML Core Configuration
SAML_ENTRYPOINT=https://idp.yourbank.com/saml/sso
SAML_ISSUER=ClientIQ-Production
SAML_CALLBACK_URL=https://clientiq.yourbank.com/api/auth/saml/acs
SAML_CERT=MIIDxTCCAq2gAwIBAgIJAKPv...base64...==
```

Optional SAML Settings

Variable	Default	Description
SAML_LOGOUT_URL	-	IdP logout URL
SAML_LOGOUT_CALLBACK_URL	-	SP logout callback
SAML_DECRYPT_KEY	-	Private key for encrypted assertions
SAML_SIGNATURE_ALGORITHM	sha256	Signature algorithm
SAML_DIGEST_ALGORITHM	sha256	Digest algorithm
SAML_CLOCK_SKEW_MS	300000	Clock skew tolerance (5 min)

Optional SAML Settings

SAML_LOGOUT_URL=https://idp.yourbank.com/saml/logout

SAML_LOGOUT_CALLBACK_URL=https://clientiq.yourbank.com/api/auth/saml/logout/callback

SAML_SIGNATURE_ALGORITHM=sha256

SAML_CLOCK_SKEW_MS=300000

Custom Attribute Mapping

Override default SAML attribute URIs:

```
SAML_ATTR_EMPLOYEE_ID=urn:oid:1.2.840.113556.1.4.1
SAML_ATTR_EMPLOYEE_NUMBER=urn:oid:1.2.840.113556.1.4.2
SAML_ATTR_FIRST_NAME=urn:oid:2.5.4.42
SAML_ATTR_LAST_NAME=urn:oid:2.5.4.4
SAML_ATTR_EMAIL=urn:oid:0.9.2342.19200300.100.1.3
SAML_ATTR_DEPARTMENT=urn:oid:2.5.4.11
SAML_ATTR_ROLE=urn:oid:1.2.840.113556.1.4.3
```

Logging Configuration

LOG_LEVEL

Required: No

Default: info

Values: error, warn, info, debug, trace

Application log verbosity.

```
LOG_LEVEL=info
```

LOG_FORMAT

Required: No

Default: json in production

Values: json, pretty

Log output format.

```
LOG_FORMAT=json
```

DEBUG

Required: No

Default: (empty)

Enable debug namespaces (Node.js debug module).

```
# Enable all debug output
DEBUG=*
```

```
# Enable specific namespaces
DEBUG=passport:*,saml:*
```

```
# Enable application-specific
DEBUG=clientiq:*
```

Security Settings

CORS_ORIGIN

Required: No

Default: Same origin only

Allowed CORS origins.

```
CORS_ORIGIN=https://clientiq.yourbank.com
```

RATE_LIMIT_WINDOW_MS

Required: No

Default: 60000 (1 minute)

Rate limiting window.

```
RATE_LIMIT_WINDOW_MS=60000
```

RATE_LIMIT_MAX_REQUESTS

Required: No

Default: 100

Maximum requests per window.

```
RATE_LIMIT_MAX_REQUESTS=100
```

Feature Flags

ENABLE_AUDIT_LOGGING

Required: No

Default: true

Enable detailed audit logging.

```
ENABLE_AUDIT_LOGGING=true
```

ENABLE_PERFORMANCE_MONITORING

Required: No

Default: false

Enable performance metrics collection.

```
ENABLE_PERFORMANCE_MONITORING=true
```

Complete Production Example


```
# =====
# ClientIQ Production Configuration
# =====

# Application
NODE_ENV=production
PORT=5000
HOST=127.0.0.1

# Database (SQL Server)
DATABASE_DIALECT=sqlserver
MSSQL_SERVER=sql-server.yourbank.local
MSSQL_DATABASE=ClientIQ
MSSQL_USER=svc_clientiq
MSSQL_PASSWORD=YourSecurePassword123!
MSSQL_PORT=1433
MSSQL_ENCRYPT=true
MSSQL_TRUST_SERVER_CERTIFICATE=false

# Connection Pool
DB_POOL_MIN=5
DB_POOL_MAX=20

# Session
SESSION_SECRET=a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6q7r8s9t0u1v2w3x4y5z6
SESSION_TIMEOUT=1800000
SESSION_COOKIE_SECURE=true

# SAML Authentication
SAML_ENTRYPOINT=https://idp.yourbank.com/saml/sso
SAML_ISSUER=ClientIQ-Production
SAML_CALLBACK_URL=https://clientiq.yourbank.com/api/auth/saml/acs
SAML_LOGOUT_URL=https://idp.yourbank.com/saml/logout
SAML_CERT=MIIDxTCCAq2gAwIBAgIJAKPv...

# Logging
LOG_LEVEL=info
LOG_FORMAT=json

# Security
CORS_ORIGIN=https://clientiq.yourbank.com
RATE_LIMIT_WINDOW_MS=60000
RATE_LIMIT_MAX_REQUESTS=100

# Features
ENABLE_AUDIT_LOGGING=true
ENABLE_PERFORMANCE_MONITORING=true
```

Validation

Startup Validation

The application validates required variables on startup:

```
[Config] Validating environment configuration...
[Config] NODE_ENV: production
[Config] DATABASE_DIALECT: sqlserver
[Config] SAML_ENTRYPOINT: configured
[Config] SESSION_SECRET: configured (64 chars)
[Config] Environment validation complete
```

Missing Variables

If required variables are missing:

```
[Config] ERROR: Missing required environment variables:  
- SAML_CERT  
- SESSION_SECRET  
[Config] Application cannot start without required configuration
```

Security Best Practices

1. **Never commit .env files** to version control
2. **Use strong, random secrets** for SESSION_SECRET
3. **Encrypt sensitive values** in transit and at rest
4. **Rotate secrets** periodically (annually minimum)
5. **Limit file permissions** on .env files
6. **Use different values** for each environment

File Permissions (Windows)

```
# Secure .env file  
$acl = Get-Acl "C:\ClientIQ\config\.env"  
$acl.SetAccessRuleProtection($true, $false)  
  
# Only SYSTEM and Administrators  
$rule = New-Object System.Security.AccessControl.FileSystemAccessRule(  
    "NETWORK SERVICE", "Read", "None", "None", "Allow")  
$acl.SetAccessRule($rule)  
$rule = New-Object System.Security.AccessControl.FileSystemAccessRule(  
    "Administrators", "FullControl", "None", "None", "Allow")  
$acl.SetAccessRule($rule)  
  
Set-Acl "C:\ClientIQ\config\.env" $acl
```

Next Steps

1. [Troubleshooting Guide](#)
2. [Return to Overview](#)